

B DAT04 — PRELIMINARY ASSESSMENT · TEST II PRACTICAL DEMONSTRATION

Botika Maligaya dataset · 45 points · 120 minutes · draw ONE task from each band

One task per sheet. Write your EVIDENCE and DECISION on this sheet and hand it in.

Name: Sanchez, Eliah Therrien E. Drawn at: 10:39 AM Finished: 11:03 AM

BAND A · TASK A1 · 8 POINTS

A single operation with one correct outcome

What did you just load?

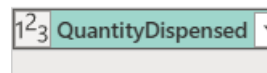
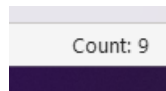
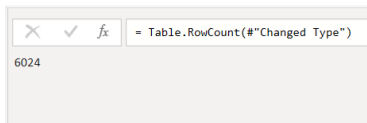
Load the main sales file. Report the number of rows, the number of columns, and the data type Power Query has assigned to QuantityDispensed. In one sentence, say why that type is a problem for a fact table.

EVIDENCE: (the counted result, visible on screen)

Number of Rows: 6024

Number of Columns: 9

Data type: Whole number



DECISION: (written, in full sentences)

The TransactionID, Branch ID, Product ID, and Senior Citizen ID or in general, all types that are in text data type is a problem for a fact table because it is unlike columns like QuantityDispensed that has a whole number data type which makes it easier for data to load.

Applied steps renamed? YES

Query name: Changed column type into a whole number

```
Table.TransformColumnTypes(#"Promoted headers", {{"TransactionID", type text}, {"Date", Int64.Type}, {"BranchID", type text}, {"ProductID", type text}, {"QuantityDispensed", Int64.Type}, {"UnitPrice", type number}, {"TotalAmount", type number}, {"PaymentMethod", type text}, {"SeniorCitizenID", type text}})
```

BAND A · TASK A2 · 8 POINTS

A single operation with one correct outcome

Prove a column is clean

Load the main sales file. Count the distinct values in PaymentMethod and list each one with its row count. State in one sentence whether this column needs cleaning.

EVIDENCE: *(the counted result, visible on screen)*

Distinct values in PaymentMethod: 4

☒ (Select All)

☒ Card

☒ Cash

☒ GCash

☒ HMO

TOTAL ROWS COUNT: 6024

Card: 790 rows

Cash: 3184 rows

GCash: 1462 rows

HMO: 588 rows

TOTAL: 6024 rows

DECISION: *(written, in full sentences)*

This column does not need cleaning as it meets all the quality requirements for a clean data and has the exact count of rows.

Applied steps renamed? YES Query name: Counted how many rows for each payment method

= Table.SelectRows("#Changed column type to whole number", each ([PaymentMethod] = "HMO"))

Name: Sanchez, Eliah Therrien E. Drawn at: 11:15 AM Finished: 11:26 AM

BAND A · TASK A3 · 8 POINTS

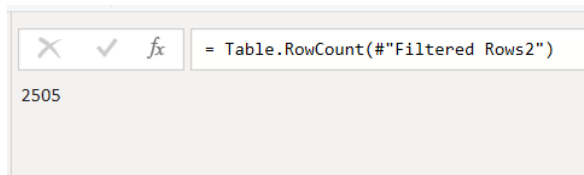
A single operation with one correct outcome

A blank that may not be a defect

Load the main sales file. Count how many rows have a blank SeniorCitizenID and express it as a percentage of all rows. State in one sentence whether this is a data quality defect.

EVIDENCE: *(the counted result, visible on screen)*

Blank: 2505



It is 41.58% of all rows.

DECISION: *(written, in full sentences)*

This is a quality defect because missing senior citizen ID could mean that the customer received a discount without it being recorded or verified.

Applied steps renamed? YES Query name: Counted how many null SeniorCitizenID

= Table.SelectRows("#Changed Type", each ([SeniorCitizenID] = null))

BAND B · TASK B1 · 10 POINTS

Two steps where the order matters

The column that will not convert

QuantityDispensed will not convert to a whole number. Repair it so the column sums. Report how many values contained a comma, and the total quantity dispensed across the file.

EVIDENCE: *(the counted result, visible on screen)*

▼	1 ² ₃ QuantityDispensed	▼	1.2
			11

QuantityDispensed is already a whole number.

DECISION: *(written, in full sentences)*

I did not have to change anything as it is already a whole number and no values contained a comma.

Applied steps renamed? NO Query name: _____

BAND B · TASK B2 · 10 POINTS

Two steps where the order matters

Append, then dedupe

Append the July 2026 sales file to the main sales query, then remove duplicate rows. Report the row count AFTER EACH STEP, and how many rows the deduplication removed.

EVIDENCE: *(the counted result, visible on screen)*

BEFORE: 54 rows

13 COLUMNS, 54 ROWS

AFTER: 49 rows

13 COLUMNS, 49 ROWS

APPLIED STEPS	
Source	⚙
Removed Duplicates	

DECISION: *(written, in full sentences)*

The deduplication removed 5 rows.

Applied steps renamed? YES

Query name: Removed all duplicates

Finished: 11:38 AM

BAND B • TASK B3 • 10 POINTS

Two steps where the order matters

More branches than exist

BranchID shows more branches than the branch list contains. Repair the formatting variants — delete nothing — then report how many distinct BranchID values remain and how many rows you repaired.

EVIDENCE: *(the counted result, visible on screen)*

TOTAL ROWS: 54

Queries [5]

- tblBranches
- SalesJu2026
- Sales
- SalesJu2026 (2)
- Append1

Table.SelectRows(Source, each ([BranchID] = "B01" or [BranchID] = "B02" or [BranchID] = "B03" or [BranchID] = "B04" or [BranchID] = "B05" or [BranchID] = "B06" or [BranchID] = "PH01" or [BranchID] = "PH02" or [BranchID] = "PH03" or [BranchID] = "PH04" or [BranchID] = "PH05" or [BranchID] = "PH06" or [BranchID] = "PH07" or [BranchID] = "PH08"))

TransactionID	Date	BranchID	Product
1	null	B01	
2	null	B02	
3	null	B03	
4	null	B04	
5	null	B05	
6	null	B06	

13 COLUMNS, 6 ROWS
Column profiling based on top 1000 rows

There seems to be two variants of the branches. Ex: B01 and PH01.

DECISION: *(written, in full sentences)*

I noticed that those branch ID that starts with B has null data. Therefore I removed 6 rows from the table while 48 remain.

Applied steps renamed? YES Query name: Showed how many rows branch ID that starts with B

BAND C · TASK C1 · 12 POINTS

A diagnostic — find it, count it, say what it costs

Products that do not exist

Using a left anti join, find every sales row whose ProductID does not appear in the product list. Count them, name the value or values, and state what happens to those rows in a report broken down by product category.

EVIDENCE: *(the counted result, visible on screen)*

6 COLUMNS, 79 ROWS

= Table.NestedJoin(Products, {"ProductID"}, SalesJul2026, {"ProductID"}, "SalesJul2026", JoinKind.LeftAnti)

DECISION: *(written, in full sentences)*

There are a lot of skips in missing product ID. Example: DRG117,....DRG120. It costed a lot of ProductID to be missing.

Applied steps renamed? NO Query name: _____

BAND C · TASK C2 · 12 POINTS

A diagnostic — find it, count it, say what it costs

Branches that do not exist

Using a left anti join, find every sales row whose BranchID does not appear in the branch list. Count them, then SEPARATE the ones that are a formatting problem from the ones that are genuinely unknown. Report both counts.

EVIDENCE: *(the counted result, visible on screen)*

☒ (Select All)

☒ B01

☒ B02

☒ B03

☒ B04

☒ B05

☒ B06

Does not appear on the branch list: 6

☒ (Select All)

☒ PH01

☒ PH02

☒ PH03

☒ PH04

☒ PH05

☒ PH06

☒ PH07

☒ PH08

Appears on the branch list: 8

= Table.NestedJoin(SalesJul2026, {"BranchID"}, tblBranches, {"BranchID"}, "tblBranches", JoinKind.LeftAnti)

DECISION: *(written, in full sentences)*

There are 6 branches that does not appear on the branch list and 8 that appears on the branch list.

Applied steps renamed? NO Query name: _____

BAND C · TASK C3 · 12 POINTS

A diagnostic — find it, count it, say what it costs

The number that does not reconcile

Compare TotalAmount against QuantityDispensed × UnitPrice. Count the rows that disagree, and SPLIT the disagreement by direction. State which direction has an innocent explanation and which does not.

EVIDENCE: *(the counted result, visible on screen)*

TotalAmount: 853,402.86

20 Sum: 853402.86

Quantity Dispensed: 3319 UnitPrice: 135,148.49

Sum: 3319	Sum: 135148.49
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Rows checked: 1

Rows disagreeing: 1

TotalAmount > Quantity × UnitPrice:

TotalAmount < Quantity × UnitPrice:

DECISION: *(written, in full sentences)*

The direction TotalAmount > Quantity x UnitPrice is has the innocent explanation and the TotalAmount < Quantity x UnitPrice does not. I used excel for the evidences as I could not figure out how to compare TotalAmount against QuantityDispensed x UnitPrice using Power BI.

Applied steps renamed? NO Query name: _____

BAND D · TASK D1 · 15 POINTS

A judgement — decide, and defend it in writing

The orders that look wrong

After repairing QuantityDispensed, run an outlier test on it. Report the upper fence and how many rows exceed it. Decide what to do with those rows — fix, exclude, escalate or accept — and defend the decision in writing.

EVIDENCE: *(the counted result, visible on screen)*

TotalAmount: 853,402.86

20 Sum: 853402.86

Quantity Dispensed: 3319 UnitPrice: 135,148.49

Sum: 3319

Sum: 135148.49

Total Amount / UnitPrice = 6.31

DECISION: *(written, in full sentences)*

I would accept the rows because the negative numbers may represent the customers returning or cancelling their purchase.

Applied steps renamed? NO Query name: _____

BAND D · TASK D2 · 15 POINTS

A judgement — decide, and defend it in writing

The negative numbers

QuantityDispensed contains negative values. Count them, state what they represent, and decide: fix, exclude, escalate or accept. Defend the decision in writing.

EVIDENCE: (the counted result, visible on screen)

= Table.SelectRows("#Changed Type", each ([QuantityDispensed] = -4 or [QuantityDispensed] = -3))

	A ^B _C TransactionID	1 ² ₃ Date	A ^B _C BranchID	A ^B _C ProductID	1 ² ₃ QuantityDispensed	1.2 UnitP
1	T004895		45698 PH08	DRG089	-3	
2	T003529		45894 PH04	DRG011	-4	
3	T000732		45671 PH03	DRG090	-4	
4	T005197		46177 PH02	DRG118	-3	
5	T001449		45967 PH04	DRG040	-3	
6	T003673		45786 PH01	DRG053	-3	
7	T002822		45756 PH04	DRG106	-4	
8	T003756		46156 PH03	DRG039	-3	

9 COLUMNS, 8 ROWS

DECISION: (written, in full sentences)

I would accept it because this may represent a return or a cancellation.

Applied steps renamed? NO Query name: _____

BAND D · TASK D3 · 15 POINTS

A judgement — decide, and defend it in writing

The month with nowhere to land

You have appended the July 2026 sales file. Test the sales dates against the calendar table. Report what you find, and decide what must be done before this model is refreshed again. Defend the decision in writing.

EVIDENCE: *(the counted result, visible on screen)*

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed column type to who...

Counted how many rows for ...

X

Appended Query

9 COLUMNS, 96 ROWS

DECISION: *(written, in full sentences)*

I could not do the required test to execute

Applied steps renamed? YES Query name: Appended Query